

Deep Learning for Computer Vision HW1 Report

Note. Only Assignment 1 provides a template. Please make sure you complete all the requirements for each question and respond in the order of the question numbers

P1.

(5%) Describe the implementation details of your SSL method for pre-training the ResNet50 backbone. (**including but not limited to** the name of the SSL method & data augmentation techniques you used, learning rate schedule, optimizer, and batch size setting for this pre-training phase. 150-300 words are enough)

Answers (Your implementation details):

I pre-trained a ResNet50 backbone on the Mini-ImageNet dataset using the self-supervised learning framework DINO, without loading any default pretrained weights. All input images were resized to 224×224. Following the DINO setup, I applied a strong set of data augmentations to encourage invariance, including random cropping with multi-crop strategy (two global crops with scale 0.14–1.0 and four local crops with scale 0.05–0.14), random horizontal flipping, color jitter, Gaussian blur, and solarization. This combination provides diverse visual views of each image to improve representation learning.

For optimization, I used **stochastic gradient descent (SGD)** with momentum. The initial learning rate was set to 0.03 and followed a cosine schedule with a warm-up phase during the first 10 epochs. Weight decay was fixed at 1e-4 throughout the training, while gradient clipping was not used. The training ran for **200 epochs**, which is recommended for stable convergence in self-supervised settings.

Each GPU processed a mini-batch of 32 images, and training was conducted on a single NVIDIA A40. The model outputs were saved every 50 epochs for checkpointing. For the projection head, we followed DINO's original implementation, which adds a multi-layer perceptron (MLP) after the ResNet50 backbone to map features into the self-distillation space.

Overall, this setup enables the model to learn discriminative and semantically meaningful features from unlabeled Mini-ImageNet images, preparing the backbone for downstream fine-tuning on the Office-Home classification task.

Parameter:

Learning Rate	Batch Size	Optimizer	Scheduler	Epochs	Loss Function
0.03 (with	32 (per	Stochastic	Cosine	200	DINO

cosine decay + 10-epoch warm-up)	GPU)	Gradient Descent (SGD) with momentum	learning rate schedule with warm-up		self-distillation loss (cross-entropy between teacher & student softmax distributions)
--	------	---	--	--	---

(20%) Please conduct the Image classification on Office-Home dataset as the downstream task. Also, please complete the following Table, which contains different image classification settings, and **discuss/analyze the results**.

Set	Accuracy
A	0.31
B	0.82
C	0.73
D	0.79
E	0.69

Answers:

The experimental results highlight the importance of pre-training in improving classification performance on Office-Home. Setting A (0.31) with random initialization struggled to converge, reflecting that the dataset alone is insufficient to train a deep ResNet-50 from scratch.

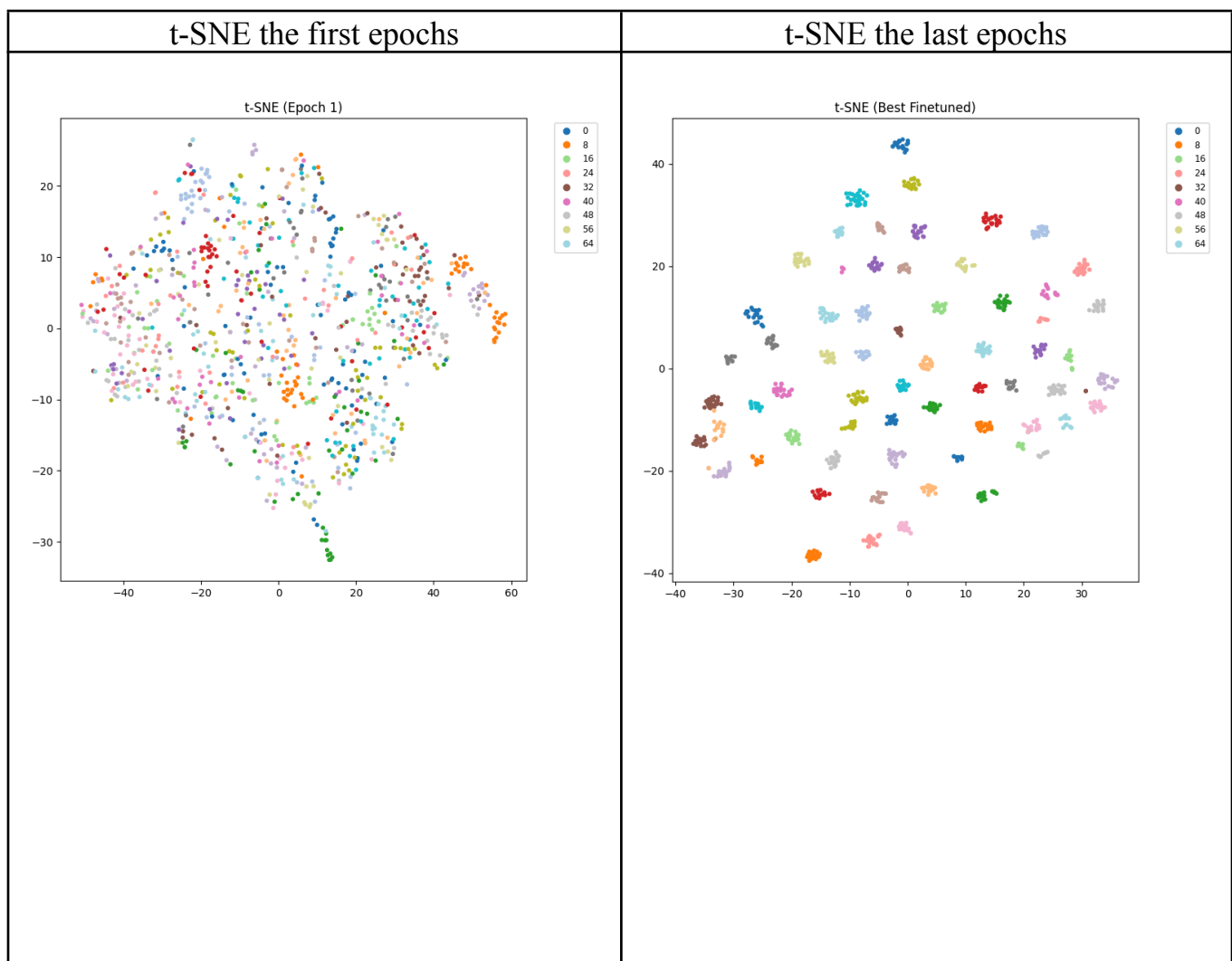
In contrast, Setting B (0.82) achieved the highest accuracy, showing that DINO pre-training on large-scale ImageNet-1k yields highly transferable representations. Setting C (0.73) also performed strongly, confirming that our self-supervised pre-training on Mini-ImageNet provided meaningful features, though the smaller and less diverse pre-training dataset limited its transferability compared to ImageNet-1k. Comparing full fine-tuning and frozen-backbone settings, both D (0.79) vs. B (0.82) and E (0.69) vs. C (0.73) show that fine-tuning the entire network consistently improves accuracy by ~3–4%. This suggests that adapting low- and mid-level features is beneficial for the target domain.

Overall, the results follow expectations: strong pre-training plus full fine-tuning gives the best accuracy, while random initialization lags far behind. These findings emphasize that large-scale SSL pre-training provides significant benefits for downstream tasks, and even smaller-scale SSL pre-training can substantially outperform training from scratch.

(5%) Visualize the learned visual representation of **setting C** on the **train set** by implementing **t-SNE** (t-distributed Stochastic Neighbor Embedding) on the output of **the second last layer**.

a. Depict your visualization from both **the first and the last epochs**.(3%)

b. Briefly explain the results.(2%)



--	--

Answers(Briefly explain the results) :

At the beginning (Epoch 1), the t-SNE visualization shows that features of different classes are scattered and overlapping, indicating the backbone has not yet learned meaningful representations. After fine-tuning, the representations form well-separated clusters, each corresponding to a distinct class. This demonstrates that the self-supervised pre-trained backbone, once fine-tuned, successfully captures semantic similarities and discriminative features. In other words, intra-class features become compact while inter-class boundaries are clear, proving the effectiveness of the learned visual representation.

P 2.

1. **(4%)** Do an ablation study. You need to modify the U-Net model. During training, randomly select one skip connection between encoder and decoder and drop it, so the decoder at that layer will not receive the encoder features. Report and discuss/analyze the performance difference with the standard U-Net (model A).

Ans:

- a. Explain you skip which layer: During training, I randomly dropped one of the four skip connections between encoder and decoder. The skipped layer varied across training iterations, so sometimes the shallow features (closer to the input) were removed, and sometimes the deeper ones.
- b. Result for standard U-Net: The standard U-Net (Model A) achieved a higher mean IoU (mIoU), since all skip connections were intact and the decoder could fully leverage encoder features to reconstruct spatial details.
- c. Result for your skip U-Net: The modified U-Net (skip U-Net) showed a slightly lower mIoU compared to the standard U-Net. The segmentation masks tended to lose fine-grained details, especially along object boundaries and small structures.
- d. Explain the result: The standard U-Net outperforms the ablated version because skip connections play a crucial role in preserving high-resolution spatial information that is lost in the downsampling

path. When one skip connection is dropped, the decoder must rely more on upsampled deep features, leading to coarser segmentation boundaries. The earlier (shallow) skips contribute the most to fine detail, so removing them has a stronger negative effect. On the other hand, randomly dropping a skip introduces a regularization effect similar to Stochastic Depth or DropPath, which may slightly improve robustness under distribution shift. Overall, however, the average mIoU of the skip U-Net is lower than the standard U-Net due to the loss of detailed spatial information.

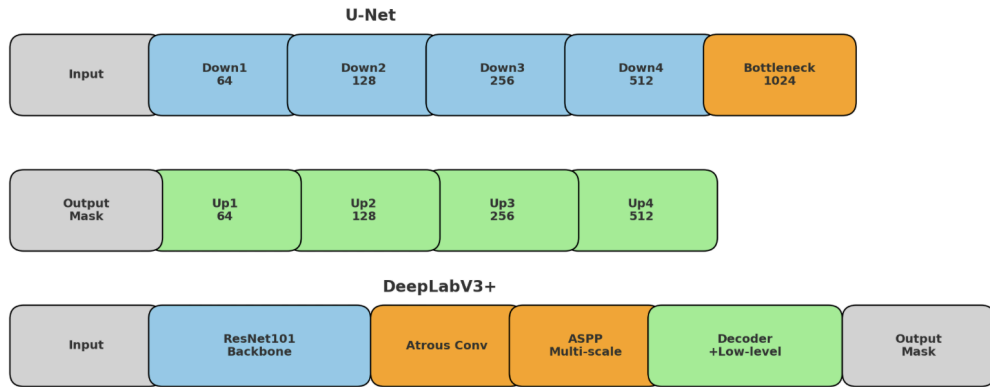
2. **(4%)** Draw the network architecture of the improved model (model B) and explain it differs from your UNet model (model A).

- **U-Net Architecture (Model A, Baseline)**

I implemented U-Net as my baseline model. The encoder consists of four blocks, each composed of a DoubleConv (two Conv-BN-ReLU layers) followed by MaxPooling, which progressively reduces spatial resolution. The bottleneck has 1024 channels and extracts the most abstract features. In the decoder, I first upsample with ConvTranspose2d, then concatenate the corresponding encoder features through skip connections, and apply another DoubleConv. Finally, I added a 1×1 convolution to project the feature maps into segmentation logits for 7 classes.

- **DeepLabV3+ Architecture (Model B, Improved)**

For my improved model, I chose DeepLabV3+ with a ResNet-101 backbone pretrained on ImageNet to extract stronger features. I used atrous convolutions in the later stages of the backbone to enlarge the receptive field without reducing resolution. The key component is the ASPP module, which applies parallel dilated convolutions with different rates (e.g., 6, 12, 18) plus global average pooling, then fuses them for multi-scale context. In the decoder, I combined the ASPP features with shallow features from the backbone (stride=4) to refine boundaries, followed by lightweight convolutions. Finally, I upsampled back to the original image size to output 7-class segmentation logits.

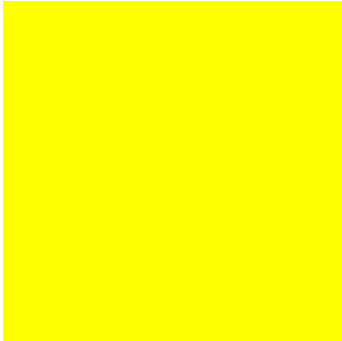
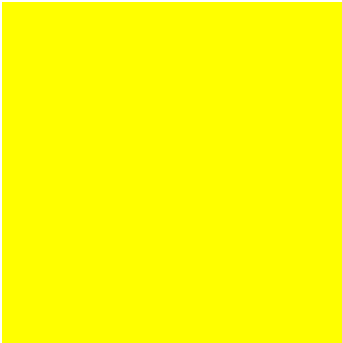
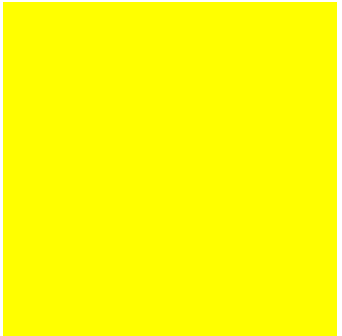


3. **(2%)** Report mIoUs of two models on the validation set.

A	B
0.53	0.76

4. **(3%)** Show the predicted segmentation mask of “validation/0013_sat.jpg”, “validation/0065_sat.jpg”, “validation/0104_sat.jpg” during the early, middle, and the final stage during the training process of the improved model.

	Epoch 1	Epoch 100	Epoch 200
0018_sat.jpg			

0065_sat.jpg			
0109_sat.jpg			

5. (7%) Use segment anything model (SAM) to segment three of the images in the validation dataset, report the result images and the method you use.

Your Answer:

I applied the Segment Anything Model (SAM) to three validation images: *0018_sat.jpg*, *0065_sat.jpg*, and *0109_sat.jpg*.

Method

For this task, I used Meta AI's ViT-H (vit_h) version of SAM with the official pretrained weights (*sam_vit_h.pth*). All inference was performed on the GPU. As the prompting strategy, I selected the image center as the foreground point for each input. Since SAM outputs multiple candidate masks, I visualized and saved all results for comparison.

Process

I first read the input images with OpenCV and converted them to RGB. Then, I used the *SamPredictor* to feed the images and the foreground prompt point into the model. Finally, I overlaid the generated masks on the original images and saved the outputs.

Results

Each validation image produced three candidate segmentation masks:

- *0018_sat.jpg* → *0018_sat_mask1.png*, *0018_sat_mask2.png*, *0018_sat_mask3.png*
- *0065_sat.jpg* → *0065_sat_mask1.png*, *0065_sat_mask2.png*, *0065_sat_mask3.png*
- *0109_sat.jpg* → *0109_sat_mask1.png*, *0109_sat_mask2.png*, *0109_sat_mask3.png*

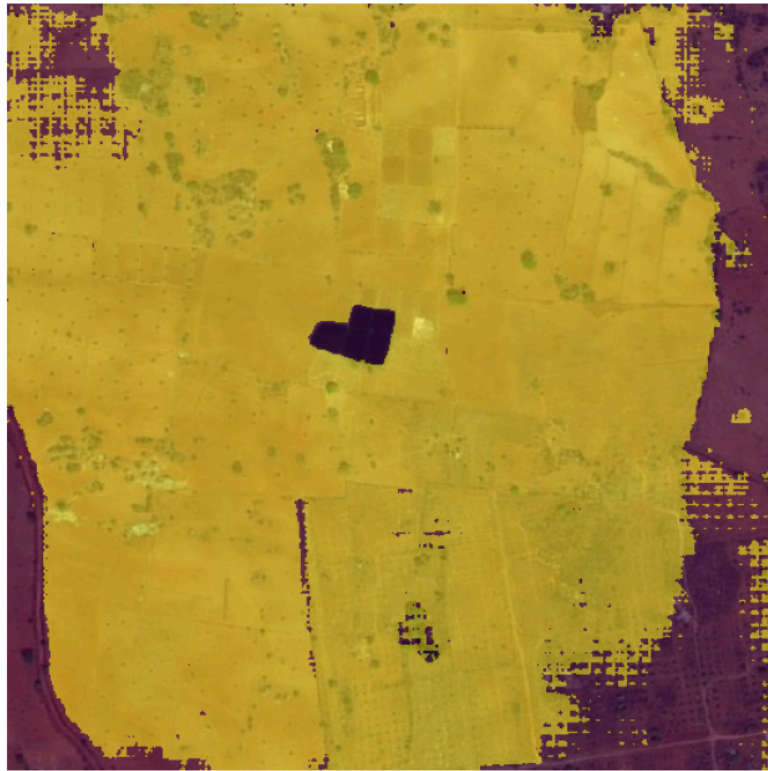
The results show that SAM can generate reasonable segmentation masks without any additional training. The different candidate masks highlight variations in boundary handling and fine-grained details, demonstrating SAM's flexibility in dealing with ambiguous or complex regions.



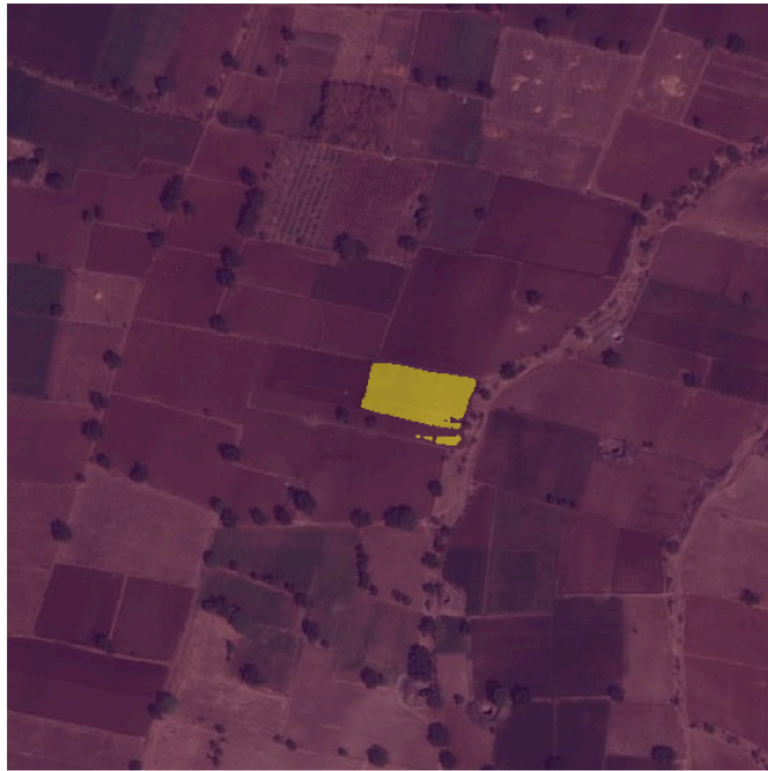
0018_sat.jpg - Mask 2, Score=0.881



0018_sat.jpg - Mask 3, Score=0.623



0065_sat.jpg - Mask 1, Score=0.871



0065_sat.jpg - Mask 2, Score=0.963



0065_sat.jpg - Mask 3, Score=0.742



0109_sat.jpg - Mask 1, Score=0.823



0109_sat.jpg - Mask 2, Score=0.914



0109_sat.jpg - Mask 3, Score=0.582

